

Zero degree Turán density without a vanishing order

Abstract

For every pair of integers $k > \ell \geq 3$, we construct a fixed k -uniform hypergraph F with $\pi_\ell(F) = 0$ and with no ℓ -vanishing order. Writing $d = k - \ell$, the construction has $7d^2 + 13d + \ell + 2$ vertices and $14d + 6$ edges. It nevertheless has an $(\ell + 1)$ -vanishing order. The proof uses finite Ramsey theory, a set-intersection estimate, and an explicit embedding argument.

1 Definitions and statement

For a k -graph H and an ℓ -set $S \subseteq V(H)$, write

$$d_H(S) = |\{e \in E(H) : S \subseteq e\}|, \quad \delta_\ell(H) = \min_{S \in \binom{V(H)}{\ell}} d_H(S).$$

The ℓ -degree Turán density of a fixed k -graph F is

$$\pi_\ell(F) = \lim_{n \rightarrow \infty} \frac{\text{ex}_\ell(n, F)}{\binom{n-\ell}{k-\ell}},$$

where $\text{ex}_\ell(n, F)$ is the largest minimum ℓ -degree of an n -vertex F -free k -graph. Copies are injective, not necessarily induced.

For an ordering $\prec$ of $V(F)$ and an edge e , let $p_e(v)$ denote the position of $v \in e$ in the increasing ordering of e . The ordering is ℓ -*vanishing* if every ℓ -set contained in two edges occupies the same set of positions in both edges. Equivalently, if $S \subseteq e \cap f$ and $|S| = \ell$, then $p_e(v) = p_f(v)$ for every $v \in S$. This is the definition in Ai, Ding, Liu and Yang [1]. Their Conjecture 1.5 asserts that $\pi_\ell(F) = 0$ implies the existence of such an ordering when $k > \ell \geq 3$.

Theorem 1. *For every $k > \ell \geq 3$, put $d = k - \ell$. There is a k -graph $F_{k,\ell}$ such that*

$$\pi_\ell(F_{k,\ell}) = 0, \quad F_{k,\ell} \text{ has no } \ell\text{-vanishing order.}$$

It has exactly $7d^2 + 13d + \ell + 2$ vertices and $14d + 6$ edges, and it has an $(\ell + 1)$ -vanishing order.

We first construct a $(d + 3)$ -graph F_d for the degree parameter 3. Adding a common set of $\ell - 3$ vertices to its edges will give $F_{k,\ell}$.

2 The construction for minimum three-degree

Fix $d \geq 1$, and put

$$q = d + 1, \quad m = 3q.$$

Take vertices $v_1, \dots, v_m$, and write

$$(a_i, b_i, c_i) = (v_{3i-2}, v_{3i-1}, v_{3i}) \quad (1 \leq i \leq q).$$

Take pairwise disjoint d -sets $X_1, \dots, X_q, Z$, all disjoint from these vertices, and distinguish one vertex $x_i \in X_i$ for each i . For every $1 \leq j \leq m-2$, take two further $(d+1)$ -sets P_j, Q_j . All the sets P_j, Q_j are mutually disjoint and disjoint from every previously specified vertex.

The edge set of F_d consists of the following edges, and no others:

$$e_i = \{a_i, b_i, c_i\} \cup X_i, \quad f_i = \{a_i, c_i, x_i\} \cup Z \quad (1 \leq i \leq q), \quad (1)$$

$$P_j \cup \{v_j, v_{j+1}\}, \quad Q_j \cup \{v_j, v_{j+2}\} \quad (1 \leq j \leq m-2), \quad (2)$$

$$Q_j \cup \{v_j, v_{j+2}\}, \quad Q_j \cup \{v_{j+1}, v_{j+2}\} \quad (1 \leq j \leq m-2). \quad (3)$$

Every edge has $d+3$ vertices. The vertex sets used above are disjoint as specified, so all displayed edges are distinct. Moreover,

$$|V(F_d)| = m + qd + d + 2(m-2)(d+1) = 7d^2 + 13d + 5,$$

$$|E(F_d)| = 2q + 4(m-2) = 14d + 6.$$

3 The ordering obstruction

Proposition 2. *The hypergraph F_d has no 3-vanishing order.*

Proof. Suppose that $\prec$ is a 3-vanishing order. First consider two edges $T \cup \{x\}$ and $T \cup \{y\}$, where $|T| = d+2 \geq 3$. For any $u \in T$, choose a 3-subset of T containing u . Its positions must agree in the two edges. Since the edges differ only by replacing x with y , this gives

$$\mathbf{1}_{\{x \prec u\}} = \mathbf{1}_{\{y \prec u\}}.$$

Thus u cannot lie strictly between x and y .

Applied to (2), this says that v_j is not between v_{j+1} and v_{j+2} . Applied to (3), it says that v_{j+2} is not between v_j and v_{j+1} . Consequently, v_{j+1} lies between v_j and v_{j+2} for every j . The successive comparisons therefore all have the same direction:

$$v_1 \prec v_2 \prec \dots \prec v_m \quad \text{or} \quad v_m \prec v_{m-1} \prec \dots \prec v_1.$$

For each i , let I_i be the open interval between a_i and c_i in this order. The intervals $I_1, \dots, I_q$ are pairwise disjoint, and $b_i \in I_i$.

The two edges e_i, f_i in (1) meet in exactly $\{a_i, c_i, x_i\}$. Hence a_i and c_i have the same positions in both edges, so the numbers of edge vertices strictly between them agree:

$$|e_i \cap I_i| = |f_i \cap I_i|.$$

Canceling the possible contribution of the common vertex x_i yields

$$|Z \cap I_i| = 1 + |(X_i \setminus \{x_i\}) \cap I_i| \geq 1. \quad (4)$$

Thus the d -set Z meets each of the $q = d+1$ pairwise disjoint intervals I_i , which is impossible. $\square$

4 The zero-density embedding

We include all quantitative estimates needed to make the embedding independent of the host order and to ensure injectivity.

Lemma 3. *Let $0 < \alpha \leq 1$, let Ω be a finite nonempty set, and put $s = \lceil 2/\alpha^2 \rceil$. If $A_1, \dots, A_s \subseteq \Omega$ satisfy $|A_i| \geq \alpha|\Omega|$, then some $i < j$ satisfy*

$$|A_i \cap A_j| \geq \frac{\alpha^2}{2} |\Omega|.$$

Proof. Let $t(\omega) = |\{i : \omega \in A_i\}|$. Cauchy–Schwarz gives $\sum_{\omega} t(\omega)^2 \geq s^2 \alpha^2 |\Omega|$, while $\sum_i |A_i| \leq s|\Omega|$. It follows that

$$\frac{1}{\binom{s}{2}} \sum_{i < j} |A_i \cap A_j| \geq \frac{s\alpha^2 - 1}{s - 1} |\Omega| \geq \frac{\alpha^2}{2} |\Omega|.$$

The last inequality follows from $(s + 1)\alpha^2 \geq 2$. $\square$

Proposition 4. *For every $d \geq 1$ and every $\varepsilon > 0$, every sufficiently large $(d + 3)$ -graph H satisfying*

$$\delta_3(H) \geq \varepsilon \binom{n - 3}{d}, \quad n = |V(H)|,$$

contains F_d . In particular, $\pi_3(F_d) = 0$.

Proof. It suffices to consider $0 < \varepsilon < 1$. In addition to $q = d + 1$ and $m = 3q$, define

$$s = \left\lceil \frac{8}{\varepsilon^2} \right\rceil, \quad r = \left\lceil \frac{2d}{\varepsilon} \right\rceil + 1, \quad M = \max\{s + 1, 3r\},$$

and put $N = |V(F_d)| = 7d^2 + 13d + 5$ and $B_0 = M + dr$. Let $R_3(M; 4)$ be the finite Ramsey number for four colors on triples. We require

$$n \geq R_3(M; 4), \quad n \geq 12(d + 1), \quad n \geq \frac{4dB_0}{\varepsilon}, \quad n \geq \frac{16(d + 1)N}{\varepsilon^2}. \quad (5)$$

All these requirements depend only on d, ε .

A set with uniformly large pair-link intersections. For distinct $u, v \in V(H)$, define

$$\mathcal{L}_{uv} = \left\{ P \in \binom{V(H)}{d+1} : P \cap \{u, v\} = \emptyset, P \cup \{u, v\} \in E(H) \right\}.$$

Double counting gives

$$(d + 1)|\mathcal{L}_{uv}| = \sum_{w \notin \{u, v\}} d_H(\{u, v, w\}) \geq (n - 2)\varepsilon \binom{n - 3}{d},$$

so

$$|\mathcal{L}_{uv}| \geq \varepsilon \binom{n - 2}{d + 1} \geq \frac{\varepsilon}{2} \binom{n}{d + 1}.$$

For the last inequality, note that at most a proportion $2(d + 1)/n$ of all $(d + 1)$ -sets meet a fixed pair, and use (5). Write

$$\Omega = \binom{V(H)}{d + 1}, \quad \tau = \frac{\varepsilon^2}{8} |\Omega|.$$

Fix an arbitrary order $<$ of the host vertices. For $u < v < w$, color $\{u, v, w\}$ by

$$(\mathbf{1}_{\{|\mathcal{L}_{uv} \cap \mathcal{L}_{uw}| \geq \tau\}}, \mathbf{1}_{\{|\mathcal{L}_{uv} \cap \mathcal{L}_{vw}| \geq \tau\}}).$$

Ramsey’s theorem yields an M -set U on which all triples receive the same color. The first entry of this color cannot be zero: otherwise, fixing the smallest vertex of U and taking s further vertices contradicts Lemma 3 with $\alpha = \varepsilon/2$. The second entry cannot be zero either, by fixing the largest vertex. Thus, for every $u < v < w$ in U ,

$$|\mathcal{L}_{uv} \cap \mathcal{L}_{uw}| \geq \tau, \quad |\mathcal{L}_{uv} \cap \mathcal{L}_{vw}| \geq \tau. \quad (6)$$

No degree condition on $H[U]$ is being asserted or used.

A common d -set for $d + 1$ triples. Choose $3r$ vertices of U , listed as

$$u_1 < v_1 < w_1 < u_2 < v_2 < w_2 < \cdots < u_r < v_r < w_r.$$

The estimates in (5) imply

$$\binom{n-3}{d} \geq \frac{3}{4} \binom{n}{d}, \quad B_0 \binom{n-1}{d-1} \leq \frac{\varepsilon}{4} \binom{n}{d}. \quad (7)$$

Indeed, the first follows because the proportion of d -sets meeting a fixed triple is at most $3d/n$, and the second follows from $\binom{n-1}{d-1}/\binom{n}{d} = d/n$.

Using the minimum three-degree and (7), choose pairwise disjoint d -sets $T_1, \dots, T_r$, all disjoint from U , such that

$$\{u_i, v_i, w_i\} \cup T_i \in E(H) \quad (1 \leq i \leq r).$$

At each step fewer than B_0 vertices are forbidden, so at least $(\varepsilon/2)\binom{n}{d}$ eligible completions remain. Choose an arbitrary $t_i \in T_i$, and set

$$W = U \cup \bigcup_{i=1}^r T_i, \quad \mathcal{Y}_i = \left\{ Z' \in \binom{V(H) \setminus W}{d} : \{u_i, w_i, t_i\} \cup Z' \in E(H) \right\}.$$

Since $|W| = B_0$, the same estimates give

$$|\mathcal{Y}_i| \geq \frac{\varepsilon}{2} \binom{n}{d}.$$

Consequently,

$$\sum_{i=1}^r |\mathcal{Y}_i| \geq \frac{r\varepsilon}{2} \binom{n}{d} > d \binom{n}{d}.$$

Some d -set Z_0 therefore belongs to at least $d + 1 = q$ of these families. Choose indices $i_1 < \cdots < i_q$ with $Z_0 \in \mathcal{Y}_{i_h}$ for all h .

Map (a_h, b_h, c_h) to $(u_{i_h}, v_{i_h}, w_{i_h})$, map X_h bijectively to T_{i_h} with $x_h \mapsto t_{i_h}$, and map Z to Z_0 . These maps are injective and already realize all edges in (1). The images $\bar{v}_1, \dots, \bar{v}_m$ of the core vertices satisfy

$$\bar{v}_1 < \bar{v}_2 < \cdots < \bar{v}_m, \quad \bar{v}_j \in U.$$

Completing the embedding. For every $1 \leq j \leq m - 2$, choose the image of P_j from

$$\mathcal{L}_{\bar{v}_j \bar{v}_{j+1}} \cap \mathcal{L}_{\bar{v}_j \bar{v}_{j+2}},$$

and the image of Q_j from

$$\mathcal{L}_{\bar{v}_j \bar{v}_{j+2}} \cap \mathcal{L}_{\bar{v}_{j+1} \bar{v}_{j+2}}.$$

Each family has at least τ members by (6). Choose these sets successively, avoiding all previously used vertices. At every step fewer than N vertices have been used, and the number of $(d+1)$ -sets meeting them is at most

$$N \binom{n-1}{d} = \frac{N(d+1)}{n} \binom{n}{d+1} \leq \frac{\varepsilon^2}{16} \binom{n}{d+1} < \tau.$$

Thus all choices succeed. This realizes (2) and (3), and all vertex images are distinct. The resulting copy is the fixed graph F_d ; only the auxiliary parameters r, M, n depend on ε .

The embedding assertion follows. Its contrapositive gives $\text{ex}_3(n, F_d) < \varepsilon \binom{n-3}{d}$ for all sufficiently large n . Since $\varepsilon > 0$ is arbitrary, $\pi_3(F_d) = 0$. $\square$

5 All degree parameters

Proof of Theorem 1. Put $d = k - \ell$, and let A be a set of $\ell - 3$ new vertices. Define

$$V(F_{k,\ell}) = A \dot{\cup} V(F_d), \quad E(F_{k,\ell}) = \{A \cup e : e \in E(F_d)\}.$$

The stated vertex and edge counts follow immediately.

Suppose first that H is an n -vertex k -graph with

$$\delta_\ell(H) \geq \varepsilon \binom{n - \ell}{d}.$$

Fix any $(\ell - 3)$ -set $A_0 \subseteq V(H)$. Its link $L_H(A_0)$ is a $(d + 3)$ -graph on $n' = n - \ell + 3$ vertices and satisfies

$$\delta_3(L_H(A_0)) \geq \varepsilon \binom{n - \ell}{d} = \varepsilon \binom{n' - 3}{d}.$$

Proposition 4 embeds F_d into this link for sufficiently large n . Adjoining A_0 gives a copy of $F_{k,\ell}$ in H . Thus $\pi_\ell(F_{k,\ell}) = 0$. This direction of the suspension principle is also an immediate instance of [1, Theorem 1.7]; the argument above proves the needed assertion directly.

If $F_{k,\ell}$ had an ℓ -vanishing order, consider any two base edges e, f and any 3-set $S \subseteq e \cap f$. The common ℓ -set $A \cup S$ has identical positions in $A \cup e$ and $A \cup f$. For each $v \in S$, deleting A subtracts the same quantity $|\{a \in A : a \prec v\}|$ from its position in both edges. Hence the restricted order on $V(F_d)$ is 3-vanishing, contrary to Proposition 2.

Finally, F_d has a 4-vanishing order: put all vertices of Z first, then all vertices of all P_j, Q_j , then all vertices of the X_i , and finally $v_1, \dots, v_m$ in that order. Within the first three groups any ordering is allowed. The only pairs of distinct edges whose intersection can have size at least four are the two edges using the same P_j or the same Q_j , and pairs f_i, f_h meeting in Z . Their common vertices have identical positions under this order. Prepending A then gives an $(\ell + 1)$ -vanishing order of $F_{k,\ell}$: any common $(\ell + 1)$ -set contains at least four base vertices, and the base positions agree while the vertices of A occupy fixed initial positions. $\square$

Remark 5. The construction is not optimized. For example, it gives a 59-vertex, 34-edge 5-graph with zero minimum-three-degree density and with no 3-vanishing order. Its common set Z has two vertices, whereas three pairwise disjoint intervals are forced to meet Z . For gap $d = 1$, smaller four-uniform examples may be used instead. The explicit $(\ell + 1)$ -vanishing order is consistent with the weaker necessary condition in [1, Theorem 1.6(a)].

References

- [1] J. Ai, L. Ding, H. Liu and H. Yang, *Vanishing orders, suspensions and zero degree Turán densities*, arXiv:2603.05973v2, 2026.